

Fixed-Span Obstructions for Overparameterized Tensor ALS and Gradient Descent

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1 Theoretical Setup

1.1 Smoothed CP target and objective

Let r, n, k be positive integers and let $q > 0$ be fixed. For arbitrary deterministic matrices $\bar{A} = [\bar{a}_1, \dots, \bar{a}_r]$, $\bar{B} = [\bar{b}_1, \dots, \bar{b}_r]$, and $\bar{C} = [\bar{c}_1, \dots, \bar{c}_r]$ in $\mathbb{R}^{n \times r}$, set $\rho = r^{-q}$. Independently draw

$$\xi_j^a, \xi_j^b, \xi_j^c \sim \mathcal{N}\left(0, \frac{\rho^2}{n} I_n\right), \quad 1 \leq j \leq r,$$

and define

$$a_j = \bar{a}_j + \xi_j^a, \quad b_j = \bar{b}_j + \xi_j^b, \quad c_j = \bar{c}_j + \xi_j^c, \quad T = \sum_{j=1}^r a_j \otimes b_j \otimes c_j.$$

No conditioning, incoherence, or random-base restriction is imposed on the deterministic matrices.

For $X = [x_1, \dots, x_k]$, $Y = [y_1, \dots, y_k]$, and $Z = [z_1, \dots, z_k]$ in $\mathbb{R}^{n \times k}$, write

$$S(X, Y, Z) = \sum_{i=1}^k x_i \otimes y_i \otimes z_i, \quad F(X, Y, Z) = \frac{1}{2} \|T - S(X, Y, Z)\|_F^2.$$

The symbol $T_{(m)}$ denotes the mode- m matricization, and $\odot$ denotes the Khatri–Rao product in one fixed standard ordering.

1.2 Method-specific fixed initialization spans

For each $M \in \{\text{cALS}, \text{cGD}\}$, independently draw $G_x^M, G_y^M, G_z^M \in \mathbb{R}^{n \times k}$ with iid $\mathcal{N}(0, 1/n)$ entries. Let

$$\mathcal{S}_M = \text{range}(G_x^M), \quad Q_M = \text{orth}(G_x^M),$$

where orth is a fixed measurable choice of an orthonormal basis on the full-column-rank set. Define

$$\mathcal{H}_M = \mathcal{S}_M \otimes \mathbb{R}^n \otimes \mathbb{R}^n, \quad P_{\mathcal{S}_M} = Q_M Q_M^\top, \quad P_{\mathcal{H}_M} = P_{\mathcal{S}_M} \otimes I_n \otimes I_n.$$

The two initialization triples are independent of one another and of the smoothing variables. Both methods use the same realization of T .

The constrained sequential alternating least-squares method initializes

$$(X_0, Y_0, Z_0) = (G_x^{\text{cALS}}, G_y^{\text{cALS}}, G_z^{\text{cALS}}).$$

At sweep t , it updates in the order X, Y, Z . With

$$K_t^x = Z_t \odot Y_t,$$

the constrained block is

$$X_{t+1} = Q_{\text{cALS}} \left[Q_{\text{cALS}}^\top T_{(1)} K_t^x ((K_t^x)^\top K_t^x)^\dagger \right].$$

It is followed by

$$K_t^y = Z_t \odot X_{t+1}, \quad Y_{t+1} = T_{(2)} K_t^y ((K_t^y)^\top K_t^y)^\dagger,$$

and

$$K_t^z = Y_{t+1} \odot X_{t+1}, \quad Z_{t+1} = T_{(3)} K_t^z ((K_t^z)^\top K_t^z)^\dagger.$$

The displayed choices are the minimum-Frobenius-norm exact block least-squares solutions, including at singular Gram matrices. The method has no restart or early stopping.

The constrained gradient method writes $X_t = Q_{\text{cGD}} C_t$ and uses

$$f_{Q_{\text{cGD}}}(C, Y, Z) = F(Q_{\text{cGD}} C, Y, Z).$$

It initializes

$$C_0 = Q_{\text{cGD}}^\top G_x^{\text{cGD}}, \quad Y_0 = G_y^{\text{cGD}}, \quad Z_0 = G_z^{\text{cGD}}.$$

At a current state $u_t = (C_t, Y_t, Z_t)$, it tests the dyadic step sizes $\eta = 2^{-j}$, $j = 0, 1, 2, \dots$, in order, and chooses the first one for which, with $V_t = u_t - \eta \nabla f_{Q_{\text{cGD}}}(u_t)$,

$$f_{Q_{\text{cGD}}}(V_t) \leq f_{Q_{\text{cGD}}}(u_t) - \frac{\eta}{2} \|\nabla f_{Q_{\text{cGD}}}(u_t)\|_F^2.$$

Here the squared gradient norm is the sum of the squared Frobenius norms of the C, Y, Z blocks. The accepted value is denoted η_t , and $u_{t+1} = V_t$, $X_{t+1} = Q_{\text{cGD}} C_{t+1}$. This method also has no restart.

For either method, set

$$S_t^M = S(X_t^M, Y_t^M, Z_t^M), \quad F_M(t) = F(X_t^M, Y_t^M, Z_t^M).$$

When $\|(I - P_{\mathcal{H}_M})T\|_F > 0$, define the fixed normalized tensor

$$W_M = \frac{(I - P_{\mathcal{H}_M})T}{\|(I - P_{\mathcal{H}_M})T\|_F}.$$

No value is assigned to W_M when the denominator is zero.

1.3 Assumptions

Assumption 1 (Fixed ambient dimension). r and n are positive integers and $n \geq 8r^{5/4}$.

Assumption 2 (Superlinear algorithmic rank). k is an integer satisfying

$$r < k \leq r^{1+c} = r^{5/4}, \quad c = \frac{1}{4}.$$

Assumption 3 (Uniform arbitrary deterministic bases). $\bar{A}, \bar{B}, \bar{C} \in \mathbb{R}^{n \times r}$ are deterministic and unrestricted. In particular, no conditioning or incoherence hypothesis is imposed, and the theorem is pointwise uniform over all such triples.

Assumption 4 (Independent Gaussian smoothing). $q > 0$ is fixed, $\rho = r^{-q}$, and all $3r$ perturbation vectors used above are mutually independent with covariance $(\rho^2/n)I_n$.

Assumption 5 (Shared target and independent starts). The two method-specific Gaussian initialization triples are mutually independent and independent of the smoothing variables. Both methods use the same realized tensor T when their events are intersected.

1.4 Formalized goal

The objective is to prove that one may take $r_0 = 1$ and that, for every admissible r, n, k and every unrestricted deterministic base triple, the joint smoothing-and-initialization law satisfies

$$\mathbb{P} \left[\bigcap_{M \in \{\text{cALS}, \text{cGD}\}} \left\{ \lim_{t \rightarrow \infty} F_M(t) \text{ exists and } \lim_{t \rightarrow \infty} F_M(t) \geq \frac{3}{8} \|T\|_F^2 \right\} \right] \geq \frac{1}{4}.$$

This is a material-partial result for the one-mode fixed-span constrained class. Replacing the fixed initialization span by the endogenous moving span of ordinary unconstrained ALS or full-variable gradient descent remains open.

2 Preliminaries

The tensor space is equipped with the Frobenius inner product $\langle A, B \rangle_F$ and its induced norm. Matricization only permutes entries, so it preserves this norm. For a first-mode subspace $\mathcal{S} \subseteq \mathbb{R}^n$, the orthogonal projector onto $\mathcal{S} \otimes \mathbb{R}^n \otimes \mathbb{R}^n$ acts on the mode-1 matricization by left multiplication with $P_{\mathcal{S}}$.

The symbols S_t^M and $F_M(t)$ retain the method superscript because the two methods use independent initialization spans while sharing the same tensor. This distinction is essential in the joint probability statement. All objective and residual bounds below use the ambient tensor Frobenius norm, and all horizon-uniform assertions refer to integer iterates $t \geq 0$.

3 Main Theorem

Theorem 3.1 (Fixed-span positive limiting objective). *Under Assumptions 1–5, set*

$$r_0 = 1, \quad \alpha = \frac{1}{4}, \quad L(r) = r^{1+\alpha} = r^{5/4}.$$

For every integer $r \geq r_0$, every integer $n \geq 8L(r)$, every integer k with $r < k \leq L(r)$, and every deterministic base triple allowed by Assumption 3, run the two constrained methods above with their independent Gaussian starts and their shared smoothed tensor T . Then, with probability at least $1/4$ over the smoothing variables and both complete initialization triples, simultaneously for $M \in \{\text{cALS}, \text{cGD}\}$,

$$\|(I - P_{\mathcal{H}_M})T\|_F^2 \geq \frac{3}{4} \|T\|_F^2, \quad F_M(t) \geq \frac{3}{8} \|T\|_F^2 \quad \text{for every } t \geq 0,$$

and the finite scalar limit satisfies

$$\lim_{t \rightarrow \infty} F_M(t) \geq \frac{3}{8} \|T\|_F^2.$$

Equivalently,

$$\mathbb{P} \left[\bigcap_{M \in \{\text{cALS}, \text{cGD}\}} \left\{ \lim_{t \rightarrow \infty} F_M(t) \text{ exists as a finite real number and } \lim_{t \rightarrow \infty} F_M(t) \geq \frac{3}{8} \|T\|_F^2 \right\} \right] \geq \frac{1}{4}.$$

All displayed constants are exact and hide no dependence on r, n, k, q, ρ, T , the deterministic bases, either initialization, the method, or the iteration index. The probability statement is pointwise uniform over the deterministic base triple, not a simultaneous event over all triples. The conclusion is only for the two displayed fixed-one-mode constrained procedures and does not extend to ordinary unconstrained ALS or full-variable gradient descent.

4 Proof Sketch

The argument separates a static geometric obstruction from the two method-specific convergence arguments. First, the Gaussian X -initialization has full column rank almost surely, and its range is a Haar-uniform k -dimensional subspace independent of the shared target. The constrained ALS formula and the coefficient parameterization of constrained GD then keep every represented tensor in the same method-specific tensor subspace; see Lemma A.1, Lemma A.2, and Proposition A.1.

Conditionally on an arbitrary realized T , Haar symmetry yields $\mathbb{E}P_{\mathcal{S}_M} = (k/n)I_n$. Hence the expected projected tensor energy is $(k/n)\|T\|_F^2$. Markov's inequality gives a probability-1/2 event for each method, and conditional independence of the two spans gives a joint probability of at least 1/4. Since $k/n \leq 1/8$, the joint event forces at least 3/4 of the target energy into each fixed orthogonal complement; this is Proposition A.3.

Fixed-span membership gives an exact orthogonal decomposition of the actual residual. It follows that every defined objective is at least one half of the fixed orthogonal-complement energy, with a time-independent normalized witness whenever that energy is nonzero; see Proposition A.4 and Lemma A.7.

For constrained ALS, compact-SVD least squares proves that all three pseudoinverse updates are exact minimum-norm block minimizers even for singular designs. Their sequential inequalities make the nonnegative objective monotone, so it has a finite scalar limit. For constrained GD, the coefficient objective is polynomial. At each finite iterate its Hessian is bounded on the current search segment, which guarantees finite dyadic Armijo acceptance and monotone objective descent. Induction gives every finite iterate and again a finite scalar limit. These conclusions are stated in Proposition A.7, Lemma A.11, Proposition A.10, and Proposition A.11.

On the joint projection event, the exact residual-energy constant 3/4 and the factor 1/2 in the objective produce the all-time constant 3/8. Passing to the two already established scalar limits loses nothing, and the tower property preserves the conditional probability 1/4. The resulting quantified statement is Proposition A.18.

The fixed-span restriction is essential to this proof. For an ordinary unconstrained first update, an explicit outside-span forcing term appears, and the orthogonal residual becomes the difference between the fixed target component and a generated tensor component. Proposition A.15 records this exact non-transfer calculation without making a success or failure claim for unconstrained dynamics.

A Proof Details

A.1 Gaussian initialization and fixed-span preservation

Lemma A.1 (Admissible dimensions and Gaussian full rank). *Under Assumptions 1, 2, and 5,*

$$\frac{k}{n} \leq \frac{1}{8} < 1,$$

and

$$\mathbb{P}(\text{rank } G_x^{\text{cALS}} = k \text{ and } \text{rank } G_x^{\text{cGD}} = k) = 1.$$

Proof. Assumptions 1 and 2 give

$$0 < k \leq r^{5/4} \leq \frac{n}{8},$$

so $k/n \leq 1/8$ and $k < n$.

We first recall the elementary polynomial fact needed below. The zero set of a nonzero real polynomial has Lebesgue measure zero. In one variable this follows because a nonzero polynomial has finitely many roots. Inductively, write a polynomial in d variables as a polynomial in its last variable, with coefficient polynomials in the first $d - 1$ variables. At least one coefficient is nonzero. The common zero set of the nonzero coefficients has measure zero by induction; outside it, the last-variable polynomial is nonzero and has finitely many roots. Fubini's theorem proves the assertion in dimension d .

Fix $M \in \{\text{cALS}, \text{cGD}\}$ and let B_M be the leading $k \times k$ block of G_x^M , which exists because $k < n$. The polynomial $B \mapsto \det B$ is not identically zero, since it equals one at I_k . The entries of B_M have a joint density, and therefore

$$\mathbb{P}(\det B_M = 0) = 0.$$

Whenever B_M is nonsingular, the k columns of G_x^M are linearly independent. Thus each matrix has column rank k almost surely. A finite union of the two null complements proves the simultaneous statement; independence is not needed for this last intersection. $\square$

Lemma A.2 (Haar initialization spans and independence). *Under Assumptions 1, 2, and 5, on the probability-one event of Lemma A.1, each*

$$\mathcal{S}_M = \text{range}(G_x^M), \quad M \in \{\text{cALS}, \text{cGD}\},$$

is Haar-uniform on the Grassmannian $\text{Gr}(k, n)$. The random elements $\mathcal{S}_{\text{cALS}}$, $\mathcal{S}_{\text{cGD}}$, and T are mutually independent. A regular conditional law of the pair of spans given $T = T_0$ may be chosen, for every T_0 , as

$$\text{Haar}_{k,n} \otimes \text{Haar}_{k,n}.$$

The corresponding subspace projectors, tensor subspaces, and tensor projectors inherit these independence properties.

Proof. Fix a method M and $U \in O(n)$. If g_i is a column of G_x^M , then

$$U g_i \sim \mathcal{N}\left(0, U(n^{-1}I_n)U^\top\right) = \mathcal{N}(0, n^{-1}I_n).$$

The columns remain independent, and hence

$$U G_x^M \stackrel{d}{=} G_x^M.$$

On the full-rank event,

$$\text{range}(UG_x^M) = U \text{range}(G_x^M) = US_M,$$

so the law of $\mathcal{S}_M$ is invariant under every orthogonal transformation.

For completeness, this invariant law is unique. Let λ be normalized Haar measure on the compact group $O(n)$ and let $\mathcal{S}_0$ be a fixed k -plane. The pushforward of λ under $U \mapsto US_0$ is invariant. If μ is any other invariant probability on $\text{Gr}(k, n)$ and h is bounded Borel, then

$$\int h(\mathcal{S}) d\mu(\mathcal{S}) = \int \int h(US) d\lambda(U) d\mu(\mathcal{S}).$$

For every $\mathcal{S} = V\mathcal{S}_0$, right invariance of λ makes the inner integral equal to $\int h(US_0) d\lambda(U)$, independently of $\mathcal{S}$. Thus μ equals the indicated pushforward. It follows that $\mathcal{S}_M$ is Haar-uniform.

Let Ξ denote the complete collection of smoothing variables used to form T . Assumption 5 makes

$$G_x^{\text{cALS}}, \quad G_x^{\text{cGD}}, \quad \Xi$$

mutually independent. The spans, represented by their measurable projectors, are measurable functions of the first two matrices, and T is a polynomial, hence measurable, function of Ξ . Measurable images preserve independence, so the two spans and T are mutually independent.

Let $\mu = \text{Haar}_{k,n} \otimes \text{Haar}_{k,n}$. For Borel sets $A \subseteq \text{Gr}(k, n)^2$ and $B \subseteq \mathbb{R}^{n \times n \times n}$,

$$\begin{aligned} & \mathbb{P}((\mathcal{S}_{\text{cALS}}, \mathcal{S}_{\text{cGD}}) \in A, T \in B) \\ &= \mu(A) \mathbb{P}(T \in B) = \int_B \mu(A) \mathbb{P}_T(dT_0). \end{aligned}$$

Therefore the constant kernel $T_0 \mapsto \mu$ is a regular conditional law for every T_0 . The projector and tensor-subspace assertions follow because they are measurable functions of the spans. $\square$

Lemma A.3 (First-factor membership lifts to tensor membership). *Let $Q \in \mathbb{R}^{n \times k}$ have orthonormal columns, set $\mathcal{S} = \text{range}(Q)$, and define*

$$\mathcal{H} = \mathcal{S} \otimes \mathbb{R}^n \otimes \mathbb{R}^n, \quad P_{\mathcal{H}} = QQ^T \otimes I_n \otimes I_n.$$

If $D \in \mathbb{R}^{k \times k}$, $X = QD$, and $Y, Z \in \mathbb{R}^{n \times k}$, then

$$S(X, Y, Z) \in \mathcal{H}, \quad P_{\mathcal{H}} S(X, Y, Z) = S(X, Y, Z), \quad (I - P_{\mathcal{H}}) S(X, Y, Z) = 0.$$

Proof. Write $D = [d_1, \dots, d_k]$. Then $x_i = Qd_i \in \mathcal{S}$, and hence

$$x_i \otimes y_i \otimes z_i \in \mathcal{S} \otimes \mathbb{R}^n \otimes \mathbb{R}^n = \mathcal{H}$$

for every i . Their sum lies in $\mathcal{H}$. Since QQ^T is the orthogonal projector onto $\mathcal{S}$, the tensor-product projector fixes every summand and therefore fixes the sum. The complementary-projector identity is equivalent. $\square$

Proposition A.1 (Exact fixed-span preservation). *Under Assumptions 1 and 2, and under Assumption 5, consider the probability-one event in Lemma A.1. Every defined iterate of either constrained method has a matrix $D_t^M \in \mathbb{R}^{k \times k}$ for which*

$$X_t^M = Q_M D_t^M.$$

Consequently, for every such index,

$$S_t^M \in \mathcal{H}_M, \quad P_{\mathcal{H}_M} S_t^M = S_t^M, \quad (I - P_{\mathcal{H}_M}) S_t^M = 0.$$

Proof. On the simultaneous full-rank event, every Q_M has k orthonormal columns spanning $\mathcal{S}_M$.

For constrained ALS, initialization gives

$$X_0^{\text{cALS}} = Q_{\text{cALS}} Q_{\text{cALS}}^\top G_x^{\text{cALS}} = Q_{\text{cALS}} D_0^{\text{cALS}}, \quad D_0^{\text{cALS}} = Q_{\text{cALS}}^\top G_x^{\text{cALS}}.$$

Whenever the next sweep is evaluated, its constrained block has the exact form

$$X_{t+1}^{\text{cALS}} = Q_{\text{cALS}} D_{t+1}^{\text{cALS}},$$

where

$$D_{t+1}^{\text{cALS}} = Q_{\text{cALS}}^\top T_{(1)} K_t^x ((K_t^x)^\top K_t^x)^\dagger.$$

Every finite real matrix has a Moore–Penrose pseudoinverse: an SVD defines it by reciprocating the nonzero singular values. Thus this factorization remains meaningful when the Gram matrix is singular or zero. The later Y - and Z -updates do not alter X_{t+1} .

For constrained GD,

$$Q_{\text{cGD}} C_0 = Q_{\text{cGD}} Q_{\text{cGD}}^\top G_x^{\text{cGD}} = G_x^{\text{cGD}} = X_0^{\text{cGD}},$$

and at every later defined index the method defines $X_t^{\text{cGD}} = Q_{\text{cGD}} C_t$. Therefore one may take $D_t^{\text{cGD}} = C_t$. Lemma A.3 now gives the three tensor-membership identities for both methods.

This argument is deliberately scoped to defined indices. It does not assume termination of future line searches; that all-index existence is proved separately below. $\square$

Proposition A.2 (Initialization and zero-leakage certificate). *Under Assumptions 1, 2, and 5, the two initialization spans are mutually independent Haar k -planes independent of T , conditionally retain the constant product-Haar law given every $T = T_0$, and every defined represented tensor of either constrained method has zero leakage from its fixed tensor subspace.*

Proof. Lemma A.1 gives simultaneous full rank with probability one and the exact arithmetic $k/n \leq 1/8 < 1$. Lemma A.2 gives the Haar and conditional independence statements. Proposition A.1 gives zero leakage at every defined index.

All boundary regimes are included. Equality $k/n = 1/8$ still gives $k < n$. The measurable orthonormalization may choose arbitrary signs or orientations, because only the span and projector are used. Singular or zero constrained ALS designs do not affect the left factor Q_{cALS} . No rank or size condition on Y_t, Z_t is required by Lemma A.3. At a currently undefined future GD index no assertion is made, while each existing index has the exact coefficient representation. The conclusions do not depend on the value of T and remain exact at $T = 0$ and in exact or zero-smoothing specializations. Finally, the mechanism is active at initialization and at the first defined transition: $X_0^M = Q_M D_0^M$, the first constrained ALS update is $Q_M D_1^M$, and the first constrained GD update changes only its coefficient matrix. $\square$

A.2 Conditional Haar projection margin

Lemma A.4 (Conditional Haar projector isotropy and tensor energy). *Under Assumptions 1, 2, and 5, and Lemma A.2, fix $M \in \{\text{cALS}, \text{cGD}\}$ and $T_0 \in \mathbb{R}^{n \times n \times n}$. Under the regular conditional law given $T = T_0$,*

$$\mathbb{E}[P_{\mathcal{S}_M} \mid T = T_0] = \frac{k}{n} I_n$$

and

$$\mathbb{E}[\|P_{\mathcal{H}_M} T\|_F^2 \mid T = T_0] = \frac{k}{n} \|T_0\|_F^2.$$

Consequently,

$$\mathbb{E}[\|P_{\mathcal{H}_M} T\|_F^2 \mid T] = \frac{k}{n} \|T\|_F^2.$$

Proof. Let

$$\bar{P}_M = \mathbb{E}[P_{\mathcal{S}_M} \mid T = T_0].$$

Haar invariance gives $U\bar{P}_MU^\top = \bar{P}_M$ for every $U \in O(n)$. Conjugation by coordinate sign flips makes every off-diagonal entry zero, and conjugation by coordinate permutations makes all diagonal entries equal. Thus $\bar{P}_M = \lambda I_n$. Since every realization is a rank- k orthogonal projector,

$$n\lambda = \text{tr}(\bar{P}_M) = \mathbb{E}[\text{tr}(P_{\mathcal{S}_M}) \mid T = T_0] = k,$$

which proves the first identity.

The tensor projector acts in the first mode, so

$$(P_{\mathcal{H}_M}T_0)_{(1)} = P_{\mathcal{S}_M}(T_0)_{(1)}.$$

Using symmetry and idempotence,

$$\begin{aligned} \|P_{\mathcal{H}_M}T_0\|_F^2 &= \|P_{\mathcal{S}_M}(T_0)_{(1)}\|_F^2 \\ &= \text{tr}\left(P_{\mathcal{S}_M}(T_0)_{(1)}(T_0)_{(1)}^\top\right). \end{aligned}$$

Taking conditional expectation yields

$$\begin{aligned} \mathbb{E}[\|P_{\mathcal{H}_M}T\|_F^2 \mid T = T_0] &= \text{tr}\left(\frac{k}{n}I_n(T_0)_{(1)}(T_0)_{(1)}^\top\right) \\ &= \frac{k}{n}\|(T_0)_{(1)}\|_F^2 = \frac{k}{n}\|T_0\|_F^2. \end{aligned}$$

The conditional kernel is valid for every T_0 , giving the final identity. $\square$

Lemma A.5 (Zero-safe per-method projection event). *Under Assumptions 1, 2, and 5, use the conclusions of Lemma A.2 and Lemma A.4. Define*

$$E_M = \left\{ \|P_{\mathcal{H}_M}T\|_F^2 \leq 2\frac{k}{n}\|T\|_F^2 \right\}.$$

Then, for every T_0 ,

$$\mathbb{P}(E_M \mid T = T_0) \geq \frac{1}{2}.$$

If $T_0 = 0$, this conditional probability is one.

Proof. Suppose first that $T_0 \neq 0$ and set $X_M = \|P_{\mathcal{H}_M}T_0\|_F^2 \geq 0$. The threshold $2(k/n)\|T_0\|_F^2$ is positive and Lemma A.4 gives

$$\mathbb{E}[X_M \mid T = T_0] = \frac{k}{n}\|T_0\|_F^2.$$

Markov's inequality, which follows from $a\mathbf{1}_{\{X_M \geq a\}} \leq X_M$ for $a > 0$, gives

$$\begin{aligned} \mathbb{P}(E_M^c \mid T = T_0) &\leq \mathbb{P}\left(X_M \geq 2\frac{k}{n}\|T_0\|_F^2 \mid T = T_0\right) \\ &\leq \frac{(k/n)\|T_0\|_F^2}{2(k/n)\|T_0\|_F^2} = \frac{1}{2}. \end{aligned}$$

If $T_0 = 0$, both sides of the inequality defining E_M are zero for every span, so E_M is sure and no division is used. $\square$

Lemma A.6 (Exact orthogonal-complement margin). *Under Assumptions 1 and 2, on E_M one has*

$$\|(I - P_{\mathcal{H}_M})T\|_F^2 \geq \left(1 - 2\frac{k}{n}\right) \|T\|_F^2 \geq \frac{3}{4} \|T\|_F^2.$$

The assertion includes $T = 0$ and the boundary $k/n = 1/8$.

Proof. Orthogonality of $P_{\mathcal{H}_M}$ gives

$$\|T\|_F^2 = \|P_{\mathcal{H}_M}T\|_F^2 + \|(I - P_{\mathcal{H}_M})T\|_F^2.$$

On E_M ,

$$\|(I - P_{\mathcal{H}_M})T\|_F^2 \geq \left(1 - 2\frac{k}{n}\right) \|T\|_F^2.$$

The assumptions give

$$\frac{k}{n} \leq \frac{r^{5/4}}{8r^{5/4}} = \frac{1}{8}, \quad 1 - 2\frac{k}{n} \geq 1 - 2\left(\frac{1}{8}\right) = \frac{3}{4}.$$

All quantities vanish when $T = 0$, and equality at $k/n = 1/8$ is allowed. $\square$

Proposition A.3 (Joint conditional and unconditional projection event). *Under Assumptions 1, 2, and 5, Lemma A.2, Lemma A.5, and Lemma A.6, the events E_{cALS} and E_{cGD} are conditionally independent given T . For every T_0 ,*

$$\begin{aligned} & \mathbb{P}(E_{\text{cALS}} \cap E_{\text{cGD}} \mid T = T_0) \\ &= \mathbb{P}(E_{\text{cALS}} \mid T = T_0) \mathbb{P}(E_{\text{cGD}} \mid T = T_0) \geq \frac{1}{4}. \end{aligned}$$

Moreover,

$$\mathbb{P}(E_{\text{cALS}} \cap E_{\text{cGD}}) \geq \frac{1}{4},$$

and on this joint event, simultaneously for both methods,

$$\|(I - P_{\mathcal{H}_M})T\|_F^2 \geq \left(1 - 2\frac{k}{n}\right) \|T\|_F^2 \geq \frac{3}{4} \|T\|_F^2.$$

Proof. Fix T_0 . The event E_M is then a measurable function only of the method-specific span, because its threshold is deterministic on this fiber. Lemma A.2 supplies the product-Haar conditional law. Hence

$$\mathbb{P}(E_{\text{cALS}} \cap E_{\text{cGD}} \mid T = T_0) = \prod_{M \in \{\text{cALS}, \text{cGD}\}} \mathbb{P}(E_M \mid T = T_0) \geq \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

This includes the $T_0 = 0$ fiber, on which each factor is one. The tower property then gives

$$\begin{aligned} \mathbb{P}(E_{\text{cALS}} \cap E_{\text{cGD}}) &= \mathbb{E}[\mathbb{P}(E_{\text{cALS}} \cap E_{\text{cGD}} \mid T)] \\ &\geq \mathbb{E}\left[\frac{1}{4}\right] = \frac{1}{4}. \end{aligned}$$

On the intersection, Lemma A.6 applies to each method. Only the two initial spans are conditionally independent; the complete trajectories need not be independent. The argument is pointwise in T_0 and uses no concentration or nondegeneracy property of the tensor law. The simultaneous rank-failure null set was already removed by Lemma A.1. $\square$

A.3 Exact fixed-witness objective floor

Proposition A.4 (Exact same-target Pythagorean floor). *Under Proposition A.1, fix one of the two methods, denote it by M , and fix an index t for which its iterate is defined. Set*

$$R_M = (I - P_{\mathcal{H}_M})T.$$

Then

$$T - S_t^M = R_M + (P_{\mathcal{H}_M}T - S_t^M), \quad R_M \perp (P_{\mathcal{H}_M}T - S_t^M),$$

and therefore

$$\|T - S_t^M\|_F^2 = \|(I - P_{\mathcal{H}_M})T\|_F^2 + \|P_{\mathcal{H}_M}T - S_t^M\|_F^2.$$

Equivalently,

$$F_M(t) = \frac{1}{2}\|R_M\|_F^2 + \frac{1}{2}\|P_{\mathcal{H}_M}T - S_t^M\|_F^2 \geq \frac{1}{2}\|(I - P_{\mathcal{H}_M})T\|_F^2.$$

The bound holds at every defined index with the same fixed residual R_M .

Proof. Write $P = P_{\mathcal{H}_M}$. Proposition A.1 gives $PS_t^M = S_t^M$, so $PT - S_t^M \in \mathcal{H}_M$. On the other hand,

$$PR_M = P(I - P)T = (P - P^2)T = 0,$$

and hence $R_M \in \mathcal{H}_M^\perp$. Thus $\langle R_M, PT - S_t^M \rangle_F = 0$, while algebraically

$$T - S_t^M = (I - P)T + PT - S_t^M = R_M + (PT - S_t^M).$$

Expanding the squared Frobenius norm proves the equality. Multiplying by the objective coefficient $1/2$ gives the exact objective decomposition, and discarding only its displayed nonnegative in-subspace term gives the floor. The residual R_M depends only on T and the fixed initialization subspace, so no recurrence, accumulated error, horizon constant, or factor-size condition is present. $\square$

Lemma A.7 (Fixed normalized witness in the nonzero-residual branch). *Under Proposition A.1 and Proposition A.4, if $\|R_M\|_F > 0$, then*

$$W_M = \frac{R_M}{\|R_M\|_F}$$

is a fixed unit tensor in $\mathcal{H}_M^\perp$ satisfying, for every defined iterate,

$$\langle W_M, S_t^M \rangle_F = 0, \quad \langle W_M, T - S_t^M \rangle_F = \langle W_M, T \rangle_F = \|R_M\|_F,$$

and hence $\|T - S_t^M\|_F \geq \|R_M\|_F$. If $\|R_M\|_F = 0$, no normalized witness is defined or used, while Proposition A.4 remains valid.

Proof. In the positive-denominator branch, normalization gives $\|W_M\|_F = 1$ and, because $R_M \in \mathcal{H}_M^\perp$, $W_M \in \mathcal{H}_M^\perp$. Proposition A.1 gives $S_t^M \in \mathcal{H}_M$, so $\langle W_M, S_t^M \rangle_F = 0$. Also

$$\begin{aligned} \langle W_M, T \rangle_F &= \langle W_M, R_M + P_{\mathcal{H}_M}T \rangle_F \\ &= \frac{\langle R_M, R_M \rangle_F}{\|R_M\|_F} = \|R_M\|_F. \end{aligned}$$

Subtracting the zero inner product with S_t^M proves the witness identity. For completeness, Cauchy–Schwarz follows by applying nonnegativity of $\|A - \lambda B\|_F^2$ at $\lambda = \langle A, B \rangle_F / \|B\|_F^2$ when $B \neq 0$; the zero case is immediate. Applying it with $A = W_M$ gives

$$\|R_M\|_F = |\langle W_M, T - S_t^M \rangle_F| \leq \|T - S_t^M\|_F.$$

Both R_M and W_M are fixed before the trajectory evolves. When $\|R_M\|_F = 0$, the quotient is not assigned a value and the unconditional Pythagorean identity is used instead. $\square$

Proposition A.5 (Boundary-complete fixed obstruction). *Under Proposition A.1, the Pythagorean floor in Proposition A.4 holds for every defined iterate, including all zero-residual and degenerate factor cases, and the normalized certificate in Lemma A.7 is used only on its positive-denominator branch.*

Proof. The two cited results prove the asserted fixed floor and witness statement. If $T = 0$, then $R_M = 0$ and the identity becomes $\|S_t^M\|_F^2 = 0 + \|S_t^M\|_F^2$. If $T \in \mathcal{H}_M$, the same zero-residual reduction applies. The algebra also remains valid at the literal full-span boundary $P_{\mathcal{H}_M} = I$, although the current rank assumptions give $k < n$. If $S_t^M = P_{\mathcal{H}_M} T$, equality holds in the objective floor. The membership producer includes initialization and every first defined update, so no burn-in is needed. A nonexistent future GD iterate is not quantified here; all-index existence is established later. Finally, zero, rank-deficient, or arbitrarily large finite factors do not change the orthogonal decomposition once the represented tensor belongs to $\mathcal{H}_M$. $\square$

A.4 Sequential constrained ALS descent and scalar convergence

Lemma A.8 (Exact CP mode-matricization identities). *For every finite $X, Y, Z \in \mathbb{R}^{n \times k}$, under the fixed Khatri–Rao ordering,*

$$S(X, Y, Z)_{(1)} = X(Z \odot Y)^\top, \quad S(X, Y, Z)_{(2)} = Y(Z \odot X)^\top, \quad S(X, Y, Z)_{(3)} = Z(Y \odot X)^\top.$$

Consequently, each matricized block least-squares objective is exactly the tensor objective F .

Proof. The i th column of $Z \odot Y$ is $z_i \otimes y_i$, while the mode-1 matricization of $x_i \otimes y_i \otimes z_i$ is $x_i(z_i \otimes y_i)^\top$. Hence

$$\sum_{i=1}^k x_i(z_i \otimes y_i)^\top = X(Z \odot Y)^\top.$$

The same coordinate calculation in modes 2 and 3 uses the ordered columns $z_i \otimes x_i$ and $y_i \otimes x_i$. Matricization permutes entries and therefore preserves the Frobenius norm, proving exact equality of the matrix and tensor residual objectives. $\square$

Lemma A.9 (Singular-design Moore–Penrose least squares). *Let $A \in \mathbb{R}^{p \times m}$ and $K \in \mathbb{R}^{m \times k}$ be finite. Then*

$$\phi(U) = \frac{1}{2} \|A - UK^\top\|_F^2, \quad U \in \mathbb{R}^{p \times k},$$

is minimized by

$$U_\star = AK(K^\top K)^\dagger,$$

and $U_\star$ is the unique minimizer of smallest Frobenius norm. If $K = U_K \Sigma V_K^\top$ is a compact SVD of positive rank and $P_K = V_K V_K^\top$, every minimizer has the form $U_\star + R(I_k - P_K)$. If $K = 0$, every U minimizes ϕ and $U_\star = 0$ is the unique minimum-norm minimizer.

Proof. First suppose $s = \text{rank}(K) > 0$, and extend U_K, V_K to orthogonal bases $[U_K \ U_0]$ of $\mathbb{R}^m$ and $[V_K \ V_0]$ of $\mathbb{R}^k$. Since $K^\top = V_K \Sigma U_K^\top$, right multiplication by $[U_K \ U_0]$ gives

$$\begin{aligned} \|A - UK^\top\|_F^2 &= \|(A - UV_K \Sigma U_K^\top)[U_K \ U_0]\|_F^2 \\ &= \|AU_K - UV_K \Sigma\|_F^2 + \|AU_0\|_F^2. \end{aligned}$$

The second term is independent of U , and the first is minimized exactly when

$$UV_K = AU_K \Sigma^{-1}.$$

Thus all minimizers are

$$U = AU_K \Sigma^{-1} V_K^\top + R(I_k - P_K).$$

The compact SVD also gives

$$\begin{aligned} K(K^\top K)^\dagger &= U_K \Sigma V_K^\top V_K \Sigma^{-2} V_K^\top \\ &= U_K \Sigma^{-1} V_K^\top, \end{aligned}$$

so the first term is $U_\star$. Orthogonality of the two right subspaces gives

$$\|U\|_F^2 = \|UV_K\|_F^2 + \|U(I_k - P_K)\|_F^2.$$

The unique smallest norm is obtained by setting the free component to zero. If $s = 0$, then $K = 0$ and $\phi(U) = \frac{1}{2}\|A\|_F^2$ for every U , while the pseudoinverse formula returns $U_\star = 0$. This proves the result at every rank without a conditioning assumption. $\square$

Lemma A.10 (Constrained orthogonal-basis block minimizer). *Let $A \in \mathbb{R}^{p \times m}$, $K \in \mathbb{R}^{m \times k}$, and $Q \in \mathbb{R}^{p \times k}$ satisfy $Q^\top Q = I_k$. Then*

$$\min_{\text{col}(X) \subseteq \text{range}(Q)} \frac{1}{2} \|A - XK^\top\|_F^2$$

has the minimum-Frobenius-norm minimizer

$$X_\star = Q(Q^\top AK)(K^\top K)^\dagger.$$

The conclusion holds for every rank of K , including $K = 0$.

Proof. Every feasible matrix has a unique representation $X = QC$, with $C = Q^\top X$. Put $P = QQ^\top$. Then

$$A - QCK^\top = (I_p - P)A + Q(Q^\top A - CK^\top).$$

The two terms are Frobenius-orthogonal because $(I_p - P)Q = 0$, and $\|QB\|_F = \|B\|_F$. Therefore

$$\|A - QCK^\top\|_F^2 = \|(I_p - P)A\|_F^2 + \|Q^\top A - CK^\top\|_F^2.$$

The first term is independent of C . Applying Lemma A.9 to $Q^\top A$ and K gives

$$C_\star = (Q^\top A)K(K^\top K)^\dagger.$$

Since $\|QC\|_F = \|C\|_F$, its minimum-norm property transfers to the feasible matrix $X_\star$. The same calculation includes the zero design. $\square$

Proposition A.6 (Exact finite constrained ALS block updates). *At any finite constrained ALS state with $Q = Q_{\text{cALS}}$, $Q^\top Q = I_k$, and $X_t \in \text{range}(Q)$, define in sequential order*

$$K_t^x = Z_t \odot Y_t, \quad K_t^y = Z_t \odot X_{t+1}, \quad K_t^z = Y_{t+1} \odot X_{t+1}.$$

The three displayed updates are finite minimum-Frobenius-norm minimizers of their current block problems, and

$$\begin{aligned} F(X_{t+1}, Y_t, Z_t) &\leq F(X_t, Y_t, Z_t), \\ F(X_{t+1}, Y_{t+1}, Z_t) &\leq F(X_{t+1}, Y_t, Z_t), \\ F(X_{t+1}, Y_{t+1}, Z_{t+1}) &\leq F(X_{t+1}, Y_{t+1}, Z_t). \end{aligned}$$

Proof. By Lemma A.8, the X -block objective is

$$\frac{1}{2} \|T_{(1)} - X(K_t^x)^\top\|_F^2.$$

Lemma A.10, with $A = T_{(1)}$ and $K = K_t^x$, gives exactly

$$X_{t+1} = Q \left[Q^\top T_{(1)} K_t^x ((K_t^x)^\top K_t^x)^\dagger \right].$$

Because X_t is feasible, the constrained minimum is no larger than its value at X_t , proving the first inequality.

With X_{t+1} fixed, Lemmas A.8 and A.9 give

$$Y_{t+1} = T_{(2)} K_t^y ((K_t^y)^\top K_t^y)^\dagger$$

as the minimum-norm Y -block minimizer, proving the second inequality. With X_{t+1}, Y_{t+1} fixed, the same lemmas give

$$Z_{t+1} = T_{(3)} K_t^z ((K_t^z)^\top K_t^z)^\dagger$$

as the minimum-norm Z -block minimizer, proving the third. Every finite Gram matrix has a finite Moore–Penrose inverse even when singular or zero, so the selected updates are finite. $\square$

Proposition A.7 (Well-defined sequential constrained ALS descent). *For every finite constrained ALS initialization on the stipulated orthonormal-basis domain, every sweep is defined, $X_t \in \text{range}(Q_{\text{cALS}})$ for every t , and*

$$F_{\text{cALS}}(t+1) \leq F_{\text{cALS}}(t).$$

More precisely, if

$$F_t^x = F(X_{t+1}, Y_t, Z_t), \quad F_t^y = F(X_{t+1}, Y_{t+1}, Z_t),$$

then

$$F_{\text{cALS}}(t+1) \leq F_t^y \leq F_t^x \leq F_{\text{cALS}}(t).$$

Proof. At initialization,

$$X_0 = G_x^{\text{cALS}} \in \text{range}(G_x^{\text{cALS}}) = \text{range}(Q_{\text{cALS}}).$$

Assume X_t, Y_t, Z_t are finite and X_t is feasible. Proposition A.6 gives a finite $X_{t+1} = Q_{\text{cALS}} C_{t+1}$, so the new X remains feasible. Then K_t^y is finite and the same proposition gives finite Y_{t+1} ; next K_t^z is finite and it gives finite Z_{t+1} . This closes the induction.

The three block inequalities are composed in the actual update order:

$$\begin{aligned} F_{\text{cALS}}(t+1) &= F(X_{t+1}, Y_{t+1}, Z_{t+1}) \\ &\leq F_t^y \leq F_t^x \leq F(X_t, Y_t, Z_t) = F_{\text{cALS}}(t). \end{aligned}$$

The Y -block uses the newly minimized X_{t+1} , and the Z -block uses both newly minimized factors. No simultaneous update is substituted. Nonuniqueness at a rank-deficient design does not change these inequalities, because the selected pseudoinverse solution is still an exact minimizer. $\square$

Lemma A.11 (Finite constrained ALS scalar limit and drop budget). *Under Proposition A.7,*

$$L_{\text{cALS}} := \lim_{t \rightarrow \infty} F_{\text{cALS}}(t) = \inf_{t \geq 0} F_{\text{cALS}}(t)$$

exists as a finite real number. For every $N \geq 1$,

$$\sum_{t=0}^{N-1} (F_{\text{cALS}}(t) - F_{\text{cALS}}(t+1)) = F_{\text{cALS}}(0) - F_{\text{cALS}}(N) \leq F_{\text{cALS}}(0).$$

Proof. The squared-loss definition gives $F_{\text{cALS}}(t) \geq 0$, and Proposition A.7 gives a nonincreasing finite sequence. Let

$$L = \inf_{t \geq 0} F_{\text{cALS}}(t), \quad 0 \leq L \leq F_{\text{cALS}}(0) < \infty.$$

For $\varepsilon > 0$, choose N with $F_{\text{cALS}}(N) < L + \varepsilon$. For every $t \geq N$,

$$L \leq F_{\text{cALS}}(t) \leq F_{\text{cALS}}(N) < L + \varepsilon,$$

which proves convergence to L . Summing the successive differences telescopes, and $F_{\text{cALS}}(N) \geq 0$ gives the stated budget. No factor convergence or stationarity conclusion is used. $\square$

Proposition A.8 (Constrained ALS scalar-limit certificate). *Under the finite constrained ALS definitions, every sweep is well-defined, each displayed block update is an exact minimum-Frobenius-norm minimizer, $F_{\text{cALS}}(t)$ is nonnegative and nonincreasing, and its finite scalar limit exists. These conclusions require no Khatri–Rao rank, conditioning, factor bound, or convergence-rate assumption.*

Proof. Lemmas A.8–A.10 identify the actual block objectives and prove the singular-design formulas. Propositions A.6 and A.7 give finite sequential descent, and Lemma A.11 gives the scalar limit.

If a design is zero, its block objective is independent of that factor and the pseudoinverse selects the zero factor, the unique minimum-norm choice. A nonzero singular design retains only its row-space component. The first sweep is covered because X_0 is feasible by construction. Tied minima and zero drops are allowed. If an objective value is zero, every later block minimum remains zero by nonnegativity. Factors or pseudoinverses may be arbitrarily large at a finite time; the result asserts finite arrays at every finite step but no uniform norm bound or continuity of the update map. $\square$

A.5 Coefficient Armijo descent and scalar convergence

Fix $Q = Q_{\text{cGD}}$ and abbreviate

$$f_Q(C, Y, Z) = F(QC, Y, Z).$$

We use the finite-dimensional product space

$$\mathcal{U} = \mathbb{R}^{k \times k} \times \mathbb{R}^{n \times k} \times \mathbb{R}^{n \times k}$$

with inner product

$$\langle (C, Y, Z), (C', Y', Z') \rangle_{\mathcal{U}} = \langle C, C' \rangle_F + \langle Y, Y' \rangle_F + \langle Z, Z' \rangle_F.$$

Its norm is exactly the block Frobenius norm used by the line search.

Lemma A.12 (Pointwise Taylor control for the coefficient loss). *If $u = (C, Y, Z) \in \mathcal{U}$ is finite and $g = \nabla f_Q(u)$, then g is finite and*

$$L(u) = \max_{0 \leq s \leq 1} \|\nabla^2 f_Q(u - sg)\|_{\text{op}} < \infty.$$

For every $0 \leq \eta \leq 1$,

$$f_Q(u - \eta g) \leq f_Q(u) - \eta \|g\|_{\mathcal{U}}^2 + \frac{L(u)\eta^2}{2} \|g\|_{\mathcal{U}}^2. \quad (\text{A.1})$$

Proof. For tensor indices a, b, c ,

$$S(QC, Y, Z)_{abc} = \sum_{i=1}^k \sum_{\ell=1}^k Q_{a\ell} C_{\ell i} Y_{bi} Z_{ci}.$$

Every entry of $T - S(QC, Y, Z)$ is therefore a polynomial in the entries of (C, Y, Z) , and

$$f_Q(C, Y, Z) = \frac{1}{2} \sum_{a,b,c} (T - S(QC, Y, Z))_{abc}^2$$

is a finite polynomial. Thus it is C^2 , and its gradient and Hessian have finite entries at every finite state.

The map $s \mapsto u - sg$ sends the compact interval $[0, 1]$ continuously to a compact segment. The function $v \mapsto \|\nabla^2 f_Q(v)\|_{\text{op}}$ is continuous, so it attains a finite maximum on that segment. This proves the assertion about $L(u)$.

For $0 \leq \eta \leq 1$, define $\phi(s) = f_Q(u - sg)$. Then

$$\phi'(0) = -\|g\|_{\mathcal{U}}^2, \quad \phi''(s) = \langle g, \nabla^2 f_Q(u - sg)g \rangle_{\mathcal{U}}.$$

Applying the fundamental theorem of calculus twice gives the integral remainder formula

$$\phi(\eta) = \phi(0) + \eta\phi'(0) + \int_0^\eta (\eta - s)\phi''(s) ds.$$

Therefore

$$\begin{aligned} f_Q(u - \eta g) &= f_Q(u) - \eta\|g\|_{\mathcal{U}}^2 + \int_0^\eta (\eta - s)\langle g, \nabla^2 f_Q(u - sg)g \rangle_{\mathcal{U}} ds \\ &\leq f_Q(u) - \eta\|g\|_{\mathcal{U}}^2 + \int_0^\eta (\eta - s)L(u)\|g\|_{\mathcal{U}}^2 ds, \end{aligned}$$

which is exactly (A.1). □

Proposition A.9 (Finite dyadic Armijo acceptance). *Under Lemma A.12, if $u \in \mathcal{U}$ is finite, $g = \nabla f_Q(u)$, and the line search tests $\eta_j = 2^{-j}$, then an acceptable trial occurs at a finite index. The first acceptable η is well-defined, $u^+ = u - \eta g$ is finite, and*

$$f_Q(u^+) \leq f_Q(u) - \frac{\eta}{2}\|g\|_{\mathcal{U}}^2. \tag{A.2}$$

Proof. If $g = 0$, the first trial $\eta_0 = 1$ leaves u unchanged and satisfies the Armijo inequality with equality. Suppose $g \neq 0$ and write $L = L(u) < \infty$. If $L = 0$, take $\eta_* = 1$. If $L > 0$, set

$$j_* = \max\{0, \lceil \log_2 L \rceil\}, \quad \eta_* = 2^{-j_*}.$$

Then $0 < \eta_* \leq 1$ and $L\eta_* \leq 1$. In both cases, Lemma A.12 gives

$$\begin{aligned} f_Q(u - \eta_* g) &\leq f_Q(u) - \eta_*\|g\|_{\mathcal{U}}^2 + \frac{L\eta_*^2}{2}\|g\|_{\mathcal{U}}^2 \\ &= f_Q(u) - \eta_* \left(1 - \frac{L\eta_*}{2}\right) \|g\|_{\mathcal{U}}^2 \\ &\leq f_Q(u) - \frac{\eta_*}{2}\|g\|_{\mathcal{U}}^2. \end{aligned}$$

Thus a trial no later than j_* is acceptable. The first acceptable trial therefore occurs after finitely many tests and obeys (A.2). Since u, g , and η are finite, so is $u^+ = u - \eta g$. □

Proposition A.10 (Well-defined coefficient GD iterates). *For the coefficient initialization above and the dyadic Armijo rule, every integer $t \geq 0$ has a finite state*

$$u_t = (C_t, Y_t, Z_t),$$

a finite accepted line-search index, and a step size $\eta_t \in (0, 1]$ for which

$$u_{t+1} = u_t - \eta_t \nabla f_Q(u_t)$$

is finite and

$$f_Q(u_{t+1}) \leq f_Q(u_t) - \frac{\eta_t}{2} \|\nabla f_Q(u_t)\|_{\mathcal{U}}^2. \quad (\text{A.3})$$

Consequently $X_t = QC_t$ and the represented tensor are defined at every finite index.

Proof. The formal initialization

$$u_0 = (Q^\top G_x^{\text{cGD}}, G_y^{\text{cGD}}, G_z^{\text{cGD}})$$

is a finite element of $\mathcal{U}$ on the probability-one domain on which the method is instantiated. If u_t is finite, Proposition A.9 produces a finite accepted trial, a finite u_{t+1} , and (A.3). Induction proves the assertion at every finite index. Since Q is fixed and finite, $X_t = QC_t$ and all inputs to the next iteration are finite as well. $\square$

Proposition A.11 (Monotone constrained GD objective and finite scalar limit). *Under Proposition A.10, for every $t \geq 0$,*

$$0 \leq F_{\text{cGD}}(t+1) \leq F_{\text{cGD}}(t) < \infty. \quad (\text{A.4})$$

For every $N \geq 1$,

$$0 \leq \sum_{t=0}^{N-1} \frac{\eta_t}{2} \|\nabla f_Q(u_t)\|_{\mathcal{U}}^2 \leq F_{\text{cGD}}(0) - F_{\text{cGD}}(N) \leq F_{\text{cGD}}(0). \quad (\text{A.5})$$

The finite scalar

$$L_{\text{cGD}} = \inf_{t \geq 0} F_{\text{cGD}}(t) \in [0, F_{\text{cGD}}(0)]$$

satisfies

$$\lim_{t \rightarrow \infty} F_{\text{cGD}}(t) = L_{\text{cGD}}. \quad (\text{A.6})$$

Proof. The coefficient and actual objectives agree exactly:

$$f_Q(u_t) = F(QC_t, Y_t, Z_t) = F_{\text{cGD}}(t). \quad (\text{A.7})$$

This finite polynomial is one half of a squared Frobenius norm and is therefore nonnegative. Substituting (A.7) into (A.3) proves (A.4). Summation telescopes and gives (A.5).

The nonincreasing sequence is bounded below by zero. For $L_{\text{cGD}} = \inf_t F_{\text{cGD}}(t)$ and $\varepsilon > 0$, choose N with $F_{\text{cGD}}(N) < L_{\text{cGD}} + \varepsilon$. For $t \geq N$,

$$L_{\text{cGD}} \leq F_{\text{cGD}}(t) \leq F_{\text{cGD}}(N) < L_{\text{cGD}} + \varepsilon,$$

which proves (A.6).

If $\nabla f_Q(u_t) = 0$, the first trial is accepted and the state is stationary. If $f_Q(u_t) = 0$, differentiability and nonnegativity force $\nabla f_Q(u_t) = 0$: otherwise, as $s \downarrow 0$,

$$f_Q(u_t - s \nabla f_Q(u_t)) = -s \|\nabla f_Q(u_t)\|_{\mathcal{U}}^2 + o(s) < 0,$$

a contradiction. Thus an exact zero objective remains zero. $\square$

Proposition A.12 (Coefficient Armijo scalar-limit certificate). *For constrained GD in coefficient variables, every dyadic line search terminates and every finite-index iterate exists. Its objective values are nonnegative and nonincreasing, and their finite scalar limit exists. No trajectory boundedness, global Lipschitz constant, uniform Hessian bound, positive all-time step-size lower bound, or parameter convergence is required.*

Proof. Lemma A.12 supplies a finite Hessian bound only on the current search segment. Proposition A.9 converts that bound into finite one-step acceptance, and Proposition A.10 closes the local finiteness condition by induction. Proposition A.11 transfers the accepted coefficient decrease to the actual tensor objective, telescopes the decreases, and proves scalar convergence.

The local numbers $L(u_t)$ may grow without bound, Armijo step sizes may tend to zero, and factor norms may be unbounded over time. None of these possibilities prevents finite acceptance at a fixed finite state. The same proof applies at $T = 0$, at zero gradient, and at zero objective, with exact preservation of the zero baseline. $\square$

A.6 Why the fixed witness does not transfer to unconstrained updates

Proposition A.13 (Exact first-update leakage for unconstrained GD and ALS). *On the probability-one full-rank domain of Lemma A.1, fix either method-specific initial triple and suppress its method superscript. Let*

$$\mathcal{S}_0 = \text{range}(X_0), \quad P_0 = P_{\mathcal{S}_0}, \quad K_0 = Z_0 \odot Y_0.$$

For one full-variable Euclidean gradient step with finite step size η_0 ,

$$X_1^{\text{uGD}} = X_0 - \eta_0 \nabla_X F(X_0, Y_0, Z_0),$$

one has

$$(I - P_0)X_1^{\text{uGD}} = \eta_0(I - P_0)T_{(1)}K_0. \quad (\text{A.8})$$

For the minimum-Frobenius-norm unconstrained ALS X -block update,

$$X_1^{\text{uALS}} = T_{(1)}K_0(K_0^\top K_0)^\dagger,$$

one has

$$(I - P_0)X_1^{\text{uALS}} = (I - P_0)T_{(1)}K_0(K_0^\top K_0)^\dagger. \quad (\text{A.9})$$

Thus first-step fixed-span preservation is equivalent to the vanishing of the corresponding displayed forcing term; it is not implied merely by $X_0 \in \mathcal{S}_0$.

Proof. Lemma A.8 gives

$$S(X, Y_0, Z_0)_{(1)} = XK_0^\top, \quad F(X, Y_0, Z_0) = \frac{1}{2}\|T_{(1)} - XK_0^\top\|_F^2.$$

For any $H \in \mathbb{R}^{n \times k}$,

$$\begin{aligned} \left. \frac{d}{d\epsilon} F(X + \epsilon H, Y_0, Z_0) \right|_{\epsilon=0} &= \langle XK_0^\top - T_{(1)}, HK_0^\top \rangle_F \\ &= \langle (XK_0^\top - T_{(1)})K_0, H \rangle_F. \end{aligned}$$

Therefore

$$\nabla_X F(X_0, Y_0, Z_0) = X_0 K_0^\top K_0 - T_{(1)} K_0,$$

and

$$X_1^{\text{uGD}} = X_0(I_k - \eta_0 K_0^\top K_0) + \eta_0 T_{(1)} K_0.$$

Since $(I - P_0)X_0 = 0$, left multiplication by $I - P_0$ proves (A.8). The calculation is unchanged if the other two factor components are updated simultaneously, because it uses the X -gradient at the common initial triple.

For ALS, let $K_0 = U\Sigma V^\top$ be an SVD. The same rowwise least-squares calculation as in Lemma A.9 gives the minimum-norm minimizer $T_{(1)}(K_0^\top)^\dagger$. The SVD yields

$$(K_0^\top)^\dagger = U\Sigma^\dagger V^\top = K_0(K_0^\top K_0)^\dagger,$$

including rank-deficient and zero K_0 . Applying $I - P_0$ proves (A.9).

The identities do not assert that either right-hand side is always nonzero. It may vanish when $K_0 = 0$, when the target contraction lies in $\mathcal{S}_0$, through rank deficiency, or by accidental cancellation. Conversely, neither unconstrained update contains the fixed left factor that enforces zero leakage in Proposition A.1. $\square$

Proposition A.14 (Exact outside-span residual decomposition). *Let P_0 be any orthogonal projector on $\mathbb{R}^n$, and define*

$$\mathcal{H}_0 = \text{range}(P_0) \otimes \mathbb{R}^n \otimes \mathbb{R}^n, \quad \Pi_0 = P_0 \otimes I_n \otimes I_n.$$

For arbitrary finite factors X_t, Y_t, Z_t , set

$$S_t = S(X_t, Y_t, Z_t), \quad R_0 = (I - \Pi_0)T, \quad \Lambda_t = (I - \Pi_0)S_t.$$

Then

$$\Lambda_t = S((I - P_0)X_t, Y_t, Z_t), \tag{A.10}$$

and

$$T - S_t = (R_0 - \Lambda_t) + (\Pi_0 T - \Pi_0 S_t), \tag{A.11}$$

is an orthogonal decomposition. Consequently,

$$\|T - S_t\|_F^2 = \|R_0 - \Lambda_t\|_F^2 + \|\Pi_0 T - \Pi_0 S_t\|_F^2, \tag{A.12}$$

and

$$F(X_t, Y_t, Z_t) = \frac{1}{2}\|R_0 - \Lambda_t\|_F^2 + \frac{1}{2}\|\Pi_0 T - \Pi_0 S_t\|_F^2. \tag{A.13}$$

If $\|R_0\|_F > 0$ and $W_0 = R_0/\|R_0\|_F$, then

$$\langle W_0, T - S_t \rangle_F = \|R_0\|_F - \langle W_0, \Lambda_t \rangle_F. \tag{A.14}$$

Proof. Writing factor columns as $x_{t,i}, y_{t,i}, z_{t,i}$ and using $I - \Pi_0 = (I - P_0) \otimes I_n \otimes I_n$ gives

$$\begin{aligned} (I - \Pi_0)S_t &= \sum_{i=1}^k ((I - P_0)x_{t,i}) \otimes y_{t,i} \otimes z_{t,i} \\ &= S((I - P_0)X_t, Y_t, Z_t), \end{aligned}$$

which proves (A.10). Nonzero leaking factor columns may cancel in the represented tensor, so no converse is asserted.

Insert $I = \Pi_0 + (I - \Pi_0)$ separately on T and S_t :

$$\begin{aligned} T - S_t &= (I - \Pi_0)T - (I - \Pi_0)S_t + \Pi_0 T - \Pi_0 S_t \\ &= (R_0 - \Lambda_t) + (\Pi_0 T - \Pi_0 S_t). \end{aligned}$$

The first summand lies in $\mathcal{H}_0^\perp$ and the second in $\mathcal{H}_0$. Their inner product vanishes, proving (A.12) and, after multiplication by $1/2$, (A.13).

In the branch $\|R_0\|_F > 0$, W_0 is a unit tensor in $\mathcal{H}_0^\perp$ and is orthogonal to the in-span residual. Therefore

$$\begin{aligned}\langle W_0, T - S_t \rangle_F &= \langle W_0, R_0 - \Lambda_t \rangle_F \\ &= \|R_0\|_F - \langle W_0, \Lambda_t \rangle_F.\end{aligned}$$

Orthogonal projection supplies no sign for the last inner product. $\square$

Proposition A.15 (Fixed-witness non-transfer certificate). *Under Propositions A.1, A.4, A.13, and A.14, the fixed-witness proof for the two constrained methods does not transfer to the ordinary unconstrained comparison dynamics from these interfaces alone. More precisely:*

(i) *For the constrained methods, $\Lambda_t = 0$, so the fixed term $\|R_0\|_F^2$ is recovered exactly.*

(ii) *For either unconstrained comparison,*

$$F(X_t, Y_t, Z_t) \geq \frac{1}{2} \|R_0 - \Lambda_t\|_F^2 \geq \frac{1}{2} (\max\{\|R_0\|_F - \|\Lambda_t\|_F, 0\})^2.$$

(iii) *If an additional estimate $\|\Lambda_t\|_F \leq \gamma \|R_0\|_F$ held for some $0 \leq \gamma < 1$, then*

$$F(X_t, Y_t, Z_t) \geq \frac{1}{2} (1 - \gamma)^2 \|R_0\|_F^2.$$

No such estimate follows from the preceding results for unconstrained iterates.

(iv) *If $R_0 - \Lambda_t \neq 0$, the endogenous tensor*

$$\widetilde{W}_t = \frac{R_0 - \Lambda_t}{\|R_0 - \Lambda_t\|_F} \in \mathcal{H}_0^\perp$$

satisfies

$$\langle \widetilde{W}_t, T - S_t \rangle_F = \|R_0 - \Lambda_t\|_F.$$

It yields a positive theorem-level floor only after a separate lower bound on its generated residual norm.

These statements show only that the constrained proof interface is absent. They do not assert that unconstrained iterates always leak, that cancellation occurs, or that unconstrained ALS or gradient descent succeeds or fails.

Proof. Proposition A.1 gives $\Lambda_t = 0$ in the constrained methods. Substitution in Proposition A.14 reproduces the exact Pythagorean floor of Proposition A.4.

For an unconstrained represented tensor, discard only the nonnegative in-span term in (A.13). The reverse triangle inequality gives

$$\|R_0 - \Lambda_t\|_F \geq \max\{\|R_0\|_F - \|\Lambda_t\|_F, 0\},$$

which proves the second assertion. Substituting the displayed conditional γ -bound proves the third. This parameter identifies one sufficient missing interface; it is not an assumption of the constrained theorem.

When $R_0 - \Lambda_t \neq 0$, both terms lie in $\mathcal{H}_0^\perp$, so the normalized difference lies there and is orthogonal to the in-span residual. Pairing it with (A.11) proves the endogenous witness identity. Its right-hand side can vanish or be arbitrarily smaller than $\|R_0\|_F$ without additional trajectory control.

Equations (A.8) and (A.9) show that zero leakage is not enforced by an unconstrained update; it requires an additional equality involving the target contraction. Accidental first-step vanishing is not an all-time invariant.

All degenerate regimes are consistent with this conclusion. If $T = 0$, then $R_0 = 0$, both displayed first-update forcing terms vanish, and no normalized fixed witness is defined. If $K_0 = 0$, both first-update leakage terms also vanish; singular K_0 is covered by the pseudoinverse. Factor leakage need not imply tensor leakage because rank-one terms can vanish or cancel. Exact complement cancellation $R_0 = \Lambda_t$ is algebraically allowed but not claimed to occur. If $R_0 = 0$ or $P_0 = I_n$, every outside-span expression reduces to zero and normalization is omitted. The endogenous witness is likewise omitted at its zero denominator. No smoothing-error term appears: the non-transfer issue is exactly the absence of an unconstrained span control in the same Frobenius geometry. $\square$

A.7 Joint probability and positive-limit assembly

Proposition A.16 (Joint-event residual and all-time objective floors). *Under Assumptions 1 and 2, Proposition A.3, Proposition A.4, Proposition A.7, and Proposition A.10, define*

$$\mathcal{E} = E_{\text{cALS}} \cap E_{\text{cGD}}.$$

On $\mathcal{E}$, for every $M \in \{\text{cALS}, \text{cGD}\}$ and every integer $t \geq 0$,

$$\|(I - P_{\mathcal{H}_M})T\|_F^2 \geq \left(1 - 2\frac{k}{n}\right) \|T\|_F^2 \geq \frac{3}{4} \|T\|_F^2 \quad (\text{A.15})$$

and

$$F_M(t) \geq \frac{3}{8} \|T\|_F^2. \quad (\text{A.16})$$

If $T \neq 0$, then W_M is well-defined, fixed in t , and certifies the same residual obstruction at every finite index. If $T = 0$, no normalized witness is used and both inequalities remain valid with zero right-hand side.

Proof. The dimension and rank assumptions give

$$\frac{k}{n} \leq \frac{r^{5/4}}{8r^{5/4}} = \frac{1}{8}. \quad (\text{A.17})$$

Proposition A.3 gives on $\mathcal{E}$, for both methods,

$$\|(I - P_{\mathcal{H}_M})T\|_F^2 \geq \left(1 - 2\frac{k}{n}\right) \|T\|_F^2.$$

Equation (A.17) gives $1 - 2k/n \geq 1 - 2(1/8) = 3/4$, proving (A.15), including equality at the boundary.

Proposition A.7 proves that every constrained ALS index exists, and Proposition A.10 proves the same for constrained GD. Therefore the defined-index scope of Proposition A.4 covers every integer $t \geq 0$. Combining its exact floor with (A.15) yields

$$\begin{aligned} F_M(t) &\geq \frac{1}{2} \|(I - P_{\mathcal{H}_M})T\|_F^2 \\ &\geq \frac{1}{2} \cdot \frac{3}{4} \|T\|_F^2 = \frac{3}{8} \|T\|_F^2. \end{aligned}$$

The only discarded quantity is the explicit nonnegative in-subspace squared residual in the exact Pythagorean identity.

If $T \neq 0$, then

$$\|(I - P_{\mathcal{H}_M})T\|_F^2 \geq \frac{3}{4}\|T\|_F^2 > 0,$$

so Lemma A.7 applies and gives

$$\langle W_M, T - S_t^M \rangle_F = \|(I - P_{\mathcal{H}_M})T\|_F \quad \text{for every } t \geq 0.$$

At $T = 0$ the quotient is omitted and the unnormalized identity gives the zero floor. $\square$

Proposition A.17 (Passage to both objective limits). *Under Proposition A.16, Lemma A.11, and Proposition A.11, the finite limits*

$$\ell_{\text{cALS}} = \lim_{t \rightarrow \infty} F_{\text{cALS}}(t), \quad \ell_{\text{cGD}} = \lim_{t \rightarrow \infty} F_{\text{cGD}}(t)$$

exist, and on $\mathcal{E}$ satisfy simultaneously

$$\ell_M \geq \frac{3}{8}\|T\|_F^2, \quad M \in \{\text{cALS}, \text{cGD}\}. \quad (\text{A.18})$$

Proof. Lemma A.11 gives existence and finiteness of the constrained ALS limit, and Proposition A.11 gives the constrained GD limit. On $\mathcal{E}$, Proposition A.16 gives

$$F_M(t) \geq c_T := \frac{3}{8}\|T\|_F^2 \quad \text{for every } t \geq 0.$$

If $\ell_M < c_T$, choose $\varepsilon = (c_T - \ell_M)/2 > 0$. Convergence would give an N such that

$$F_M(t) < \ell_M + \varepsilon = \frac{\ell_M + c_T}{2} < c_T \quad \text{for every } t \geq N,$$

contradicting the all-time floor. Thus (A.18) holds for each method. No limit is interchanged with expectation, probability, or optimization. $\square$

Proposition A.18 (Quantitative specialization and constrained positive-limit theorem). *Fix $q > 0$ and set*

$$\alpha = \frac{1}{4}, \quad L(r) = r^{1+\alpha} = r^{5/4}, \quad r_0 = 1.$$

Under Assumptions 1–5, for every integer $r \geq r_0$, every integer $n \geq 8L(r)$, every integer $r < k \leq L(r)$, and every unrestricted deterministic $\bar{A}, \bar{B}, \bar{C} \in \mathbb{R}^{n \times r}$, draw

$$a_j = \bar{a}_j + \xi_j^a, \quad b_j = \bar{b}_j + \xi_j^b, \quad c_j = \bar{c}_j + \xi_j^c,$$

where all perturbations are independent and

$$\xi_j^a, \xi_j^b, \xi_j^c \sim \mathcal{N}\left(0, \frac{r^{-2q}}{n} I_n\right),$$

and set $T = \sum_{j=1}^r a_j \otimes b_j \otimes c_j$. Independently draw the two complete Gaussian initialization triples as in Assumption 5, use the same T for both methods, and run exactly the constrained procedures in the theoretical setup. Then

$$\mathbb{P}\left[\bigcap_{M \in \{\text{cALS}, \text{cGD}\}} \left\{ \lim_{t \rightarrow \infty} F_M(t) \text{ exists as a finite real number and } \lim_{t \rightarrow \infty} F_M(t) \geq \frac{3}{8}\|T\|_F^2 \right\}\right] \geq \frac{1}{4}. \quad (\text{A.19})$$

On the underlying probability-1/4 event, the residual-energy and all-time objective bounds (A.15)–(A.16) also hold for both methods. The constants hide no dependence on r, n, k, q, ρ, T , the deterministic bases, either initialization, the method, the iteration, factor norms, design conditioning, or line-search history. The conclusion is pointwise uniform over each fixed deterministic base triple and concerns only the displayed constrained procedures; it makes no assertion for ordinary unconstrained ALS or full-variable gradient descent.

Proof. Fix admissible q, r, n, k and a deterministic base triple, and fix an arbitrary possible tensor value T_0 . Proposition A.3 gives under the regular conditional initialization law

$$\mathbb{P}(\mathcal{E} \mid T = T_0) \geq \frac{1}{4}. \quad (\text{A.20})$$

The probability-one Gaussian full-rank domain can be intersected with $\mathcal{E}$ without changing this bound. The constrained ALS and GD existence results are pathwise and introduce no further random event.

On $\mathcal{E}$, Proposition A.17 gives both limit conclusions in (A.19). Therefore, for every T_0 ,

$$\begin{aligned} & \mathbb{P} \left[\bigcap_{M \in \{\text{cALS}, \text{cGD}\}} \left\{ \lim_{t \rightarrow \infty} F_M(t) \text{ exists and } \lim_{t \rightarrow \infty} F_M(t) \geq \frac{3}{8} \|T\|_F^2 \right\} \middle| T = T_0 \right] \\ & \geq \mathbb{P}(\mathcal{E} \mid T = T_0) \geq \frac{1}{4}. \end{aligned} \quad (\text{A.21})$$

This is pointwise in T_0 and does not use a favorable smoothing event or independence of the complete trajectories.

Averaging (A.21) over the smoothing law and using the tower property gives

$$\begin{aligned} & \mathbb{P} \left[\bigcap_{M \in \{\text{cALS}, \text{cGD}\}} \left\{ \lim_{t \rightarrow \infty} F_M(t) \text{ exists and } \lim_{t \rightarrow \infty} F_M(t) \geq \frac{3}{8} \|T\|_F^2 \right\} \right] \\ & = \mathbb{E}_T \left[\mathbb{P} \left(\bigcap_M \left\{ \lim_{t \rightarrow \infty} F_M(t) \text{ exists and } \lim_{t \rightarrow \infty} F_M(t) \geq \frac{3}{8} \|T\|_F^2 \right\} \middle| T \right) \right] \\ & \geq \mathbb{E}_T \left[\frac{1}{4} \right] = \frac{1}{4}. \end{aligned}$$

Because the conditional argument holds for every T_0 , no property of the deterministic bases and no smoothing concentration, conditioning, or incoherence event is used. The same numerical constants apply after fixing each deterministic base triple; no simultaneous event over all triples is claimed.

The complete quantitative specialization is explicit:

$$k \leq L(r) = r^{5/4}, \quad n \geq 8L(r), \quad \frac{k}{n} \leq \frac{1}{8},$$

and therefore

$$1 - 2\frac{k}{n} \geq \frac{3}{4}, \quad \frac{1}{2} \left(1 - 2\frac{k}{n} \right) \geq \frac{3}{8}.$$

The two conditional method probabilities satisfy $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$, and the tower conversion preserves this constant. No term is hidden or absorbed. The arithmetic is valid for every positive integer r , so $r_0 = 1$ is admissible; if the rank window is empty for a small r , the universal assertion over admissible k is vacuous. The identity $L(r) = r^{1+\alpha}$ gives $\alpha = 1/4$.

At $T = 0$, both projection events are sure, both scalar limits exist, and the required lower bound is exactly zero; no normalized witness is needed. For nonzero T on $\mathcal{E}$, both witness denominators are positive. At $k/n = 1/8$, the constants are exactly $3/4$ and $3/8$. Zero block drops, singular constrained ALS designs, and zero constrained GD gradients are already included in the scalar-limit results. The shared target prevents an unconditional independence assertion, but conditional independence and the tower property suffice.

Finally, Proposition A.15 shows that ordinary unconstrained updates introduce the generated complement component Λ_t and do not inherit the fixed-witness interface. Thus the theorem neither states nor implies an unconstrained positive-limit result. $\square$

A.8 Proof of the Main Theorem

Proof of Theorem 3.1. Proposition A.18 verifies the choices $r_0 = 1$, $\alpha = 1/4$, and $L(r) = r^{5/4}$; all dimension and rank conditions; the exact $3/4$ residual-energy and $3/8$ objective constants; the conditional product probability and tower conversion giving confidence $1/4$; the existence of both finite scalar limits; and pointwise uniformity over unrestricted deterministic bases. Its event also contains the all-time floors from Proposition A.16. These conclusions are precisely those asserted in Theorem 3.1. The constrained-only scope follows from Proposition A.15, which exposes the uncontrolled outside-span term for ordinary unconstrained updates without making an asymptotic claim about those dynamics. $\square$